Departamento de Engenharia Informática e de Sistemas

	Operations Research 2015/2016	
Date: 22/01/2016	**Exam – 1st Call**	**Duration: 2 hours**

Note: Present **all the calculations** that you perform and appropriately **justify** your answers.

1. Consider the following problem:

“The cosmetics company FOREVERYOUNG intends to carry out a television show to promote its products, providing to this end, a budget of **40000** euros. This show should last for **30** minutes and should include the performance of a magician and of a dance group. The company wants that at least **4** minutes are devoted to commercials of their products. However, the TV channel requires that the time dedicated to commercials is, at the maximum, **25%** of the time allocated to both the performances of the magical and dance group. In turn, the magician is not willing to perform more than **15** minutes, being the remaining time dedicated to the dance group and commercials. The magician's performance costs **2000** euros per minute, the dance group’s acting costs **1000** euros per minute, and the cost of the commercials is also of **1000** euros per minute. Based on audience studies, it is estimated that for every minute the magician acts, **2000** viewers turn to the referred TV channel, for every minute the group dances, **4000** new viewers come up, and for each minute of commercials, **1000** viewers turn off the TV or change to another channel. To attain the greatest success in terms of products’ promotion, the company intends to have a television show that maximizes the audience (number of viewers)."

To help FOREVERYOUNG, formulate the problem in terms of a linear programming model, indicating the meaning of the decision variables and objective function.

2. Consider the following linear programming problem:

$$
\begin{array}{l}
\textit{Maximize} \;\; z = -9x_1 + 3x_2 \\
\text{subject to} \\
\qquad -2x_1 + x_2 \le 4 \\
\qquad 3x_1 + 4x_2 \ge 12 \\
\qquad x_1 + x_2 \le 10 \\
\qquad x_1 \ge 0 ,\; x_2 \ge 0
\end{array}
$$

a) Solve the problem by the Simplex method using the “Two Phases” technique;

b) What could you conclude in **a)** if the value of **z** at the end of the first phase was negative? If the “Big M” technique had been used instead of the “Two Phases”, in which situation could you take a similar conclusion?

Scores: 1 – 3,5 points 2 – 5,5 points 3 – 5,5 points 4 – 5,5 points

3. Consider now the following linear programming problem:

Maximize $z = 4x_1 + 3x_2$
subject to

$$-x_1 + x_2 \leq 3$$
$$2x_1 + x_2 \geq 6$$
$$2x_1 + 2x_2 \geq 4$$
$$x_1 \geq 0 \text{ , } x_2 \textit{ free}$$

a) Solve it by the graphical method;

b) Formulate de dual problem that corresponds to the one presented above;

c) Without solving the dual problem, is it possible to take a conclusion about its optimal solution without having the optimal table of the primal? Justify.

4. The company UNIVERSE OF GREEDY of Coimbra has three small production units (**P1**, **P2** and **P3**) specialized in Santa Clara pastries manufacturing. Usually the pastries produced in these three units are then transported to four distribution centers (**C1**, **C2**, **C3** and **C4**) that are responsible for their delivering to the many pastry shops around the country. For transportation, the pastries are packed in boxes of 100 units each. In **P1**, **P2** and **P3,** there are daily available **5**, **5** e **6** boxes of pastries, respectively. In turn, **C1**, **C2**, **C3** and **C4**, daily require **4**, **4,** **6** and **2** boxes of pastries, respectively. Follows the table with the unitary transportation costs (in euros per box) from each production unit to each distribution centre:

	C1	**C2**	**C3**	**C4**
P1	4	1	4	9
P2	2	6	6	0
P3	1	9	2	10

a) Determine an initial basic feasible solution for the problem using the Matrix Minimum method;

b) Starting from the solution obtained in a), solve the problem by the transportation method;

c) Explain the reason why the variables of the dual problem (u_i and v_j) used in the resolution of **b)** may assume negative values.

Scores: 1 – 3,5 points 2 – 5,5 points 3 – 5,5 points 4 – 5,5 points